SUNDAY PROJECT 6

AI StudyMate

Smart Study Planner & Learning Assistant

A responsive, browser-based study organization app

Abstract

AI StudyMate is a student-focused web application that brings study planning, task tracking, note-taking, flashcard practice, quizzes, and progress monitoring into one workspace. Students can organize subjects and topics, set daily study targets, track focus sessions, and review quiz results through a dashboard. Built with HTML, CSS, and JavaScript, the app stores information in the browser using localStorage so saved data remains available after a page refresh in the same browser. The current version provides local study support; it does not connect to an external AI service.

Problem statement

Students often manage assignments, revision plans, notes, and self-testing across separate tools or informal reminders. This can make deadlines and progress difficult to see, reduce consistency, and make it harder to know what to study next. AI StudyMate addresses this organization problem by combining core study workflows in a single, accessible web app.

Project objectives

- Provide a simple place to plan subjects, topics, target dates, and daily study hours.
- Make it easy to create tasks, track completion, and identify pending work.
- Support active recall with saved notes and question-and-answer flashcards.
- Offer a quick multiple-choice knowledge check with an immediate score.
- Show study time, task completion, and quiz performance in a progress dashboard.
- Persist data locally and offer a responsive interface with dark mode.

Intended users & technology

Intended users: students who want a lightweight personal tool for planning and reviewing their studies.

Technology: HTML for structure, CSS for responsive styling and themes, and vanilla JavaScript for interactions, charts, timer logic, and localStorage persistence.

Core features

The app is organized around a practical study cycle: plan, do, review, and reflect.

1. Smart study planner

Add subjects, topics, study dates, and daily hour targets. The dashboard displays each subject's target and tracked study time for today.

2. Task manager

Create tasks with optional subject and due date, mark them complete, and remove tasks that are no longer needed. Pending and completed counts appear in the dashboard.

3. Notes & flashcards

Save short notes or create question-and-answer flashcards. Navigate between cards and click a card to reveal its answer.

4. Quiz mode

Answer a built-in set of multiple-choice questions and see the score immediately. Quiz attempts are saved for progress tracking.

5. Progress dashboard

Review completed tasks, tracked time, latest and average quiz scores, and visual charts for recent study sessions and quiz attempts.

6. Focus timer & personalization

Run a focus session and associate it with a subject. Switch between light and dark themes; the layout adapts to smaller screens.

Typical user workflow

- Add a subject with topics, a target study date, and a daily hour goal.
- Break the subject into tasks and add due dates where useful.
- Start a focus session, choose a subject, and finish the session to record study time.
- Save notes and create flashcards to review key concepts.
- Take the quiz to practise recall and receive an instant result.
- Check the dashboard to review tasks, study time, and quiz progress; refresh the page and continue later.

Functional requirements

- The app shall allow users to create and remove subjects, tasks, notes, and flashcards.
- The app shall allow tasks to be marked complete and provide a study timer.
- The app shall calculate quiz scores and display progress summaries and charts.
- The app shall save and restore supported data using browser localStorage.

Quality attributes

- Usability: clear navigation, readable labels, and short forms support quick everyday use.
- Responsive design: the layout adjusts for desktop and mobile screen sizes.
- Accessibility basics: semantic form controls, visible labels, and accessible names on key buttons.
- Performance: the app is a single static HTML file and needs no server or package installation.
- Privacy: study records are stored in the user's browser and are not sent to a backend service.

Data storage & scope

StudyMate saves subjects, tasks, notes, flashcards, completed focus sessions, quiz scores, and theme preference in localStorage. Data is tied to the browser and device used to open the file. Clearing browser site data or moving to another browser may remove or leave the saved data behind. The current project has no user accounts, cloud synchronization, external database, or AI API integration.

Limitations

- Quiz content is a small, fixed set of sample questions rather than a subject-specific question bank.
- The app does not generate AI explanations, personalized plans, or questions; the AI name describes the project concept, not a connected AI feature.
- Progress is limited to data recorded in the current browser; there is no export, backup, or cross-device sync feature.
- Tracked time is based on completed timer sessions and should be treated as a personal estimate.

Future enhancements

- Add an optional AI API integration for study suggestions, summaries, and question generation, with secure server-side handling of API credentials.
- Allow users to customize quizzes by subject and difficulty, and review explanations after each answer.
- Add data export/import, cloud synchronization, and optional accounts.
- Add reminders, calendar views, accessibility refinements, and automated tests.

Conclusion

AI StudyMate demonstrates how a lightweight front-end application can support practical study habits by joining planning, focused work, active recall, quizzes, and progress tracking. Its single-file format makes it easy to run and demonstrate, while its modular features provide a foundation for future enhancements.